

Topic 7 - Infrastructure as a Service (IaaS) Providers

Exploring Features, Technologies, and Leading
Providers

Introduction to IaaS

- Infrastructure as a Service (IaaS) is a form of cloud computing that provides virtualized computing resources over the internet. With IaaS, businesses can rent servers, storage, and networking infrastructure on-demand, rather than owning and maintaining physical data centers. It provides maximum control over operating systems, storage, and deployed applications, making it ideal for system administrators and developers.
- This model lays the foundation for understanding how modern cloud infrastructures are structured and consumed, leading into a comparison with other cloud service models.

IaaS in Cloud Models

- IaaS is one of the three main cloud service models: IaaS, PaaS, and SaaS. It offers the highest level of flexibility and management control.
- Understanding IaaS's position helps us appreciate the unique features that make it suitable for specific use cases, which we will now explore.

Key Features of IaaS



On-demand Self-Service: Users can provision resources as needed.



Resource Pooling: Providers pool computing resources to serve multiple clients.



Rapid Elasticity: Scale resources up or down quickly.



Measured Service: Pay only for what you use. These features enable agility and cost-efficiency.



Elasticity and Scalability

- Elasticity refers to the ability of a system to automatically scale in or out depending on demand. For example, during a major sales event, an e-commerce site can automatically add servers to handle increased traffic, and reduce them once traffic returns to normal—ensuring cost optimization and performance stability.
 - To enable this kind of flexibility, IaaS depends heavily on virtualization.
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Virtualization Technology

- IaaS heavily relies on virtualization to abstract hardware resources. Virtual machines (VMs) are created on physical servers using hypervisors such as VMware ESXi, Xen, KVM, or Hyper-V. This enables multiple users to share the same physical hardware securely and efficiently, reducing hardware costs and improving scalability.
 - Since IaaS infrastructure spans the globe, let's now look at the importance of where this infrastructure is physically located.
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Geographic Presence

- Most IaaS providers have a global network of data centers. These are grouped into geographic regions and availability zones. The wider the presence, the more options a business has to place data closer to users—reducing latency and meeting data residency laws.
 - But having global data centers alone isn't enough—strategic placement plays a crucial role, as we'll now see.
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Importance of Data Center Location

- Choosing a data center location is crucial for performance and compliance. For instance, a company targeting Asian customers might choose a Singapore or Tokyo region. Data locality also affects legal compliance, e.g., GDPR compliance in Europe.
 - Once a location is selected, users need methods to interact with and manage their infrastructure.
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User Interfaces and Access to Servers

- Web-based dashboards (e.g., AWS Management Console)
 - Command-line tools (e.g., Azure CLI, AWS CLI)
 - RESTful APIs and SDKs for automation This variety ensures accessibility for both beginners and advanced users.
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Interfaces in Practice

- For example, DevOps teams often use APIs or tools like Terraform to automatically create and manage infrastructure as code. This automation improves consistency and reduces human error in large deployments.
 - Let's now consider how some businesses plan ahead and save costs through capacity reservation.
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Advanced Reservation of Capacity

- IaaS providers often offer reserved instances at a lower cost compared to on-demand instances. This is useful when you have predictable workloads. For instance, a business with a steady monthly traffic pattern can reserve capacity in advance and save up to 70%.
 - To see how this works in practice, consider a real-world academic use case.
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Real-World Example of Reservation

- Imagine a university hosting an exam portal every semester. Since they know the exact dates of high usage, they can reserve capacity in advance, ensuring both cost savings and availability.
 - But what happens when usage patterns are unpredictable? That's where auto scaling and load balancing come in.
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Auto Scaling and Load Balancing

- Auto Scaling automatically adjusts the number of VM instances to meet demand. Load Balancing distributes traffic evenly across instances. Together, they ensure high availability, fault tolerance, and cost efficiency during variable load conditions.
 - This is especially crucial for applications facing sudden traffic spikes, like e-commerce websites.
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E-commerce Example

- During flash sales, a fashion website may receive traffic spikes. Auto scaling adds more servers to handle the surge, and the load balancer routes users across these servers to prevent crashes, ensuring a smooth customer experience.
 - To guarantee performance, these services are backed by formal agreements known as SLAs.
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Understanding SLAs

- Service-Level Agreements define the expected level of service, typically including uptime (e.g., 99.99%), response times, and support levels. SLAs build trust and set accountability between provider and customer. Breaching SLAs may result in service credits.
 - The implications of these agreements are significant—especially for mission-critical applications.
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SLA Implications

- If a provider promises 99.99% uptime, this means only about 52 minutes of downtime per year is acceptable. Applications requiring high availability, like online banking, must evaluate SLAs critically before choosing a provider.
 - Underpinning these performance commitments is the technology that manages VMs—hypervisors.
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Hypervisors Overview

Hypervisors manage VMs and are classified into:

- Type 1: Bare-metal (e.g., Xen, VMware ESXi)
- Type 2: Hosted (e.g., VirtualBox) Type 1 hypervisors are more secure and commonly used in data centers. AWS uses a customized version of Xen and Nitro.

Hypervisors enable the use of multiple operating systems.



Operating System Choices

- Users can install any OS, such as Ubuntu, CentOS, Red Hat, or Windows Server. This flexibility supports a wide variety of use cases—from hosting websites on Linux to running legacy enterprise apps on Windows.
 - Now that we understand the core technology, let's examine the major players offering these services.
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Introduction to IaaS Providers

Several key players dominate the IaaS space, each with unique strengths:

- Amazon Web Services (AWS)
 - Flexiscale
 - Joyent
 - Rackspace We'll now explore each in more detail to compare what they bring to the table.
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Amazon Web Services (AWS)

- AWS is the largest and most widely used IaaS provider. It offers a vast array of services beyond compute and storage, including AI, IoT, and big data analytics. AWS supports global infrastructure with 30+ geographic regions.
- Let's take a closer look at one of AWS's flagship IaaS offerings—Amazon EC2.



Amazon EC2 Features

Elastic Compute Cloud (EC2)
allows you to launch
virtual servers on-demand.
Features include:

- Variety of instance types
(General, Compute, GPU)
- Elastic IPs
- AMIs (Amazon Machine
Images)
- Integration with Auto
Scaling, ELB, CloudWatch



Amazon
EC2



EC2 Use Cases

- A startup can launch a web app with minimal upfront cost.
- Research teams can run machine learning models on spot instances.
- Enterprises can deploy scalable microservices architecture.

Beyond AWS, there are other notable providers with unique features worth examining.

Flexiscale Overview

Flexiscale was among the first cloud IaaS platforms. It emphasized ease of use, GUI-based management, and real-time provisioning. It laid the groundwork for future European cloud innovations.



Joyent Overview

Joyent provides SmartOS-based virtual machines optimized for real-time apps. Known for supporting Node.js early, Joyent is now part of Samsung and focuses on cloud-native workloads and high-performance computing.



Rackspace Cloud Servers

- Rackspace specializes in managed cloud services. It helps businesses deploy and manage cloud infrastructure on platforms like AWS, Azure, and Google Cloud. Their focus is on customer support, making them ideal for companies with limited IT staff.





Comparing Providers

Feature	AWS	Flexiscale	Joyent	Rackspace
Global Reach	High	Low	Moderate	High
Support	Good	Moderate	Developer-focused	High-touch
Pricing Flexibility	High	Medium	Medium	Low



Choosing the Right IaaS

- Cost predictability
 - Performance and availability needs
 - Compliance requirements
 - In-house expertise A healthcare startup might choose AWS for scalability, while a small firm might pick Rackspace for its support.
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Real-World Applications of IaaS

- Education: Virtual labs, student VMs
 - E-commerce: Scalable web stores
 - Media: Video rendering farms
 - Government: Secure data processing
- These use cases highlight the flexibility of IaaS across sectors.
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Challenges with IaaS

- Cost overruns without monitoring
 - Data security risks if misconfigured
 - Vendor lock-in
 - Complexity in setup and scaling These should be planned for during adoption.
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